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**Rivalry, Capacity, and GEO Activity:
Evidence from a Staggered Difference-in-Differences Design**

By

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Abstract

The purpose of this research is to explore the effects of geopolitical competition on national behavior in geostationary orbit (GEO) in the context of yearly launches into GEO. Using the country-year unit of analysis, we first observe a positive relationship between rivalry pressure and GEO behavior. In order to progress past mere descriptive findings, a difference-in-difference model with staggered treatment assignment is then used to test our hypothesis of a causal link between rivalry and increased GEO activity. Using both two-way fixed effects as well as the Sun-Abraham interaction-weighted specification, we observe a statistically significant increase in GEO behavior after the shock of heightened rivalry pressure. This effect is particularly pronounced in nations with existing capabilities both within the aerospace industry as well as space exploration more generally, meaning that rivalry pressure does not produce GEO capability out of thin air but serves to amplify activities where such capability already exists. Robustness tests including placebo tests, alternative treatments, leave-one-group-out tests, and pre-trend diagnostics consistently reaffirm the key finding. Additional analyses including Cox proportional hazard models show that rivalry explains very little beyond the effects of existing capability on GEO activity.

1 Introduction

Why do some countries respond to rival activity through increased involvement in space activities, whereas others continue to lack initial success entering lucrative orbits? There is much discussion in the policy community about such dynamics; the underlying assumption is that a rival does the deed first, and everyone else tries to follow suit. While the explanation seems appealing, particularly in the domain of outer space where satellites facilitate communications, reconnaissance, strategic coordination, and national prestige, the story is probably too simplistic. It takes considerable time and effort to develop capabilities enabling states to conduct space activities. Access to GEO orbit implies the existence of necessary funds, launch capacities, industrial infrastructure, engineering talent, and organizational know-how.

According to this paper, rivalry and capacity should be treated as analytically separable entities. On one hand, rivalry generates incentives to act in certain ways. On the other hand, capacity determines whether or not these incentives can be transformed into actions. The distinction is important since the focal topic of this paper is the *intensive margin* of orbital behavior: how many times in a year does a country establish itself in GEO orbit, and how does that number react to a shock in the surrounding strategic environment? By analyzing the intensive margin, we address questions concerning how far rivalry affects escalation among countries that already have access to advanced technologies—the very context in which rival pressure should manifest in a causal effect.

My empirical approach involves two key steps. First, in an attempt to demonstrate the basic relationship of interest, I conduct an OLS estimation with year fixed effects. In addition, I consider a set of robustness checks including interaction-weighted regression (Sun–Abraham estimator). Second, after documenting the descriptive association with OLS, I apply the primary design: a staggered DID in which treatment is defined as a substantial jump in rival pressure. My analysis

includes both basic TWFE DID as well as event study estimations. Again, I use interaction-weighted DID from [Sun and Abraham \(2021\)](#) in order to correct negative weighting biases.

The results suggest that a parsimonious interpretation can be reached here. In the OLS framework, rivalry pressure exhibits a positive correlation with annual GEO entries, although the effect diminishes when controlling for capacity. In the main DID framework, a substantial rival pressure jump leads to a moderate but significant increase in GEO entry intensity. Diagnostics indicate that there are no pre-trends differentials in the near term, and the result based on interaction-weighted DID is basically indistinguishable from that of the TWFE benchmark. The heterogeneity analysis provides the most substantive insights: the rivalry pressure effect is largely restricted to countries with prior orbital activity and stronger economies. An additional analysis in Cox’s framework in the appendix reveals that on the extensive margin of *first* GEO entry, rivalry loses its explanatory power.

This paper is important for several reasons. First, much of the discussion in the arms race literature implies that rivalry is an automatic driver of imitation. However, my evidence suggests that rivalry indeed has an impact but only if a country has relevant capacity to begin with. Second, the paper introduces new insights into the problem of space security, which currently receives a great deal of attention due to Russia’s and China’s militarization of space.

2 Rivalry, Security Competition, and Capacity Constraints

This study represents a confluence of two streams of academic literature, one on the traditional security-dilemma/arms race approach to international relations and the other on technological diffusion/policy adoption in comparative politics/political economy. Both are important because they allow us to build an argument that leads to some very particular hypotheses regarding rivalries and space.

2.1 The Security Dilemma and Space Competition

The analytical foundation of the theoretical framework is the security dilemma. In an anarchic international system, the pursuit of one’s own security can generate insecurity for other states through a self-reinforcing process of competitive armament, leading to unintended escalation even among non-aggressive actors ([Jervis, 1978](#)). Particularly salient for the space domain is Jervis’s offense-defense paradigm, which suggests that, in circumstances where the offense-defense ratio tilts toward the attacker, the security dilemma problem will be particularly acute due to the physical exposure, dual use, and vulnerability of satellites, which cannot effectively defend themselves after an attack.

In applying this theory to space, we predict that states will react to observed growth in competitor space capabilities by developing their own orbit capabilities. Existing research highlights this insight in various ways. For example, [Harrison et al. \(2017\)](#) identify how the blurring of boundaries between military and civil uses of space generates escalatory risk. [Townsend \(2020\)](#) theorize

about the "orbital security dilemma," showing how difficulty in distinguishing between offense and defense in space operations exacerbates misperception. [Lele \(2019\)](#) analyze the India-China space rivalry as an illustration of security-dilemma processes in a dyadic relationship. From the systemic perspective, [Pekkanen and Blount \(2024\)](#)'s handbook offers an extensive survey of contemporary issues in space security, including questions of deterrence, norm-building, and escalation control. An even wider-ranging survey is provided in the multi-volume handbook of [Schrogl et al. \(2015\)](#), which focuses specifically on how national space policies, space technology applications, and space activities develop alongside the changing international security context, emphasizing that space security emerges through the interplay of strategic interests and technological-industrial capabilities.

Empirical research provides support for intensifying competition. [Weeden and Samson \(2024\)](#) describe development of counterspace capabilities by fourteen different countries, noting how investment in technologies posing threats to space assets has expanded far beyond the original superpowers. [Lewis \(2019\)](#) find that possession of satellites affects state behavior in international crises, indicating that possession of space assets carries strategic value above and beyond technical considerations. [Moltz \(2019\)](#) offer a historical analysis, tracing cycles of military restraint and competition beginning with the emergence of outer space and highlighting how norms of governance played a moderating role in otherwise purely competitive processes.

While the literature reviewed thus far provides ample support for the existence of incentives for space expansion as part of great power rivalries, one missing ingredient of the existing theoretical frameworks is that of *the capacity constraint*. The security dilemma theory predicts that states will respond to strategic incentives, but not all states have the ability to do so, particularly in a technically challenging area like space.

2.2 Technology Diffusion, Adoption, and Capacity

A second body of literature attempts to fill this gap. The innovation adoption or technology diffusion literature, introduced by Rogers (2003) in its foundational work, recognizes adoption as a process influenced by both external and internal factors. One key idea is that adoption is endogenous, requiring certain resources, institutions, experiences with other technologies, and access to complementary inputs.

In comparative politics and international political economy, this approach has been used to explain variations in policy and technology adoption between nations. For example, Berry (1990) uses event-history analysis to study policy adoption at the state level, highlighting how both internal factors like fiscal stability and the political environment, in addition to external influences from neighboring states and federal governments, can affect the policy process. Similarly, Simmons (2004) has applied this method to prove that international economic liberalization entails policy diffusion through competition, learning, and similarity of cultures among states. The common assertion of both papers that external factors can affect policy adoption only under certain internal conditions is in line with our thesis.

In the realm of technology, Comin et al. (2010) observe a notable lag in diffusion in the adoption

of advanced technologies among countries, estimated to be around 45 years. Using rigorous empirical techniques, they provide strong evidence in favor of the effects of human capital, institutional quality, trade, and prior adoption of related technologies on adoption lags, underscoring the role of capacity variables as main determinants, rather than controls.

In the realm of space policy, the findings of the above-mentioned literature lead us to distinguish between the intensive and extensive margins of adoption. While external competition may incentivize the participation of technologically capable states in certain activities, initially engaging in costly technological domains, such as geostationary orbit (GEO), appears to hinge much more heavily on a state’s capacity, such as established technological infrastructure, engineers, and the related institutional procedure.

2.3 Mechanisms and Hypotheses

Combined, the insights from these two literatures suggest a positive relation between rivalry and a country’s involvement in space-related activities. However, there are few attempts to explicate the mechanism behind the effect of rivalry on space activities. Here we make the underlying mechanism precise and express it as a set of testable hypotheses.

Mechanism 1: Signaling and deterrence

The first path through which rivalry affects a state’s spatial behavior is that rivalry indicates the shift in the strategic environment surrounding orbital deployment. Objects operating in geostationary orbit enable multiple military and commercial purposes including communications, missile early warning, signal intelligence gathering, and secure command and control. Due to their broad angular coverage and stationary position, geostationary orbit is especially conducive to wide territorial coverage. Therefore, when a state senses that its strategic rivals have grown their geostationary presence, it will be motivated to increase its activities as well, since higher rival capacity reduces the informational and communications utility provided by its orbital assets (Schrogl 2015; Harrison 2017). Even more importantly, a spike in geostationary activity by the strategic rival implies a build-up of capabilities for counterspace operations and, consequently, a threat posed to its satellites (Schrogl 2015; Pekkanen 2024).

Mechanism 2: Prestige, legitimation, and audience costs

Second, rivalry could have a more symbolic effect. For decades, space programs have been symbols of a state’s technical prowess (Moltz, 2019). An expansion of rival GEO capabilities could impose political costs—audience costs—on decision-makers for failing to respond. The legislature, allies, or domestic aerospace constituencies might all pressure civilian and military leaderships to show that the state is not being left behind. This mechanism does not, unlike the first, demand that the additional GEO satellite provides any particular military gain; what is required is simply that the state participate in the process. Again, prestige considerations will likely interact in practice with

strategic concerns because dual-use satellites qualify as both (Schrogl et al., 2015).

Mechanism 3: The timing of procurement and momentum of bureaucracy

The third mechanism, which can be considered relatively straightforward, works through internal procedures. It has been noted that GEO satellites have an average period of about five to ten years from the moment of the decision to create a new satellite until its launch; therefore, the range of quick responses that could take place in reaction to rivalry pressure would necessarily be limited to the missions that are already underway. Therefore, the only possible way to react to the changes in rivalry would be to shift priority toward specific missions within the existing launch queue. Rivalry shocks would then be able to reset the priorities, increase the political importance of space activity in the defense and foreign policies agenda, attract special funding, and push some of the missions forward in the queue. The direct implication from this mechanism is rather obvious: only countries with an existing project queue that could be accelerated have such opportunities.

Why capacity constraints bind tighter on entry than growth

The three aforementioned processes also predict a significant difference between first entry into geostationary orbit (GEO) and later growth of GEO activities once first entry occurs. Initial entry requires more than the simple reallocation of resources across projects already underway; it necessitates the creation of an entire set of facilities needed for the development of a project: manufacturing facilities, launching capabilities, orbital slots, tracking facilities, and other associated components. Fixed costs play a role, and not signaling effects, in determining first entry. Consistent with models of technology transfer between countries, outside pressure will induce technology adoption in technologically capable nations; nevertheless, outside pressure cannot supersede technology limitations without such capabilities.

Hypotheses

The mechanisms discussed above suggest three empirical hypotheses that structure the remainder of the paper:

- H1 (intensive-margin response)** Controlling for country and year fixed effects, a one-unit change in the level of rivalry is followed by a rise in the intensive margin of GEO activity, as given by $\log(1 + n_{it}^{\text{GEO}})$, where n_{it}^{GEO} is the number of GEO entries reported from country i in year t .
- H2 (capacity-conditioning)** This conditional relationship in H1 exhibits heterogeneity depending on whether the country had the necessary capabilities prior to the shock. Specifically, only countries that were more economically developed and engaged in space activities experience a significant increase in GEO entry after the shock. The effect size for other countries was negligible.

H3 (extensive-margin null) Rivalry adds little explanatory power to the decision to make *first* GEO entry. Rather, capability-related indicators such as industrial share of GDP and population will be driving the hazard of the first GEO entry event.

H1 is tested with the baseline TWFE DID model and the Sun–Abraham event-study in Section 4. H2 is tested through capacity-stratified DID specifications in Section 5. H3 is evaluated with a supplementary Cox proportional-hazards model, reported in Appendix E, and provides a theoretically motivated cross-check on the intensive-margin results reported in the main text.

3 Data and Empirical Strategy

This study employs a panel dataset of countries per year, spanning 176 countries from 1963 to 2024, which is created through the merging of four main datasets: satellite information of the world, rivalries dataset, democracy index across countries, and standard economic data. This current segment contains the methodology of the panel formation and variable selection, along with the research design.

3.1 Data Sources and Variable Construction

3.1.1 Space activity data.

All satellite and launch data are derived from the General Catalog of Artificial Space Objects (GCAT) maintained by Jonathan C. McDowell at the Harvard-Smithsonian Center for Astrophysics (McDowell 2020). GCAT produces two datasets relevant for this paper. The first, called geotab, includes all objects which have reached the geostationary orbital belt, and consists of 1,927 observations with information on orbital type (GEO/ID, GEO/S, GEO/I, GEO/D, and their sub-categories), date of launch, and current orbital characteristics. The second file, named launchlog, is an exhaustive log of launches, covering 28,662 launch missions from 1957 to early 2026, containing information on the satellite owner, the owner’s state, the launch vehicle, and launch outcome.

From these raw datasets, I construct several country-year variables. The main dependent variable, n_{it}^{GEO} , measures the number of new GEO entries attributed to country i in year t , where attribution is based on the satellite owner’s state code. In the dataset, this variable corresponds to `n_geo_entries`. I also define A_{it}^{GEO} , corresponding to `geo_entry_any`, as a binary indicator equal to one if country i records any GEO entry in year t . In addition, I calculate five-year cumulative owner payload counts, P_{it}^{5yr} , corresponding to `payloads_5yr_owner`, and the country’s corresponding share of global GEO payloads, S_{it}^{5yr} , corresponding to `payload_share_5yr_owner`. The first-entry year, F_i^{GEO} , corresponding to `first_entry_year`, records the first year in which country i registers a GEO entry and is used to define the failure event in the supplementary Cox model reported in Appendix E. Altogether, 36 countries in the dataset have made at least one GEO entry, beginning with the United States in 1963 and ending with New Zealand in 2022.

3.1.2 Data on rivalry.

Geopolitical rivalry is measured through the Strategic Rivalry dataset compiled by [Colaresi et al. \(2008\)](#). Specifically, this dataset lists pairs of states engaged in strategic rivalry—a continuous competition based on positioning or geographic dispute—as well as dating the rivalry periods for each rivalry dyad. Country-country associations are mapped to the panel using Correlates of War (COW) country codes.

Let C_{it} , corresponding to `rival_count_active`, denote the number of active strategic rivals for country i in year t . The key rival-pressure variable is R_{it} , corresponding to `rivals_share_5yr_owner`. It measures the share of all global geostationary payloads acquired over the previous five years by the strategic rivals of country i . The variable thus captures whether a country’s strategic rivals are establishing a presence in geostationary orbit. Over the sample period, 102 out of 176 countries in the panel experience one or more active rivalries, with a maximum of eight simultaneously active rivals. For treatment construction, I express this measure in percentage points as $R_{it}^{\text{pct}} = 100 \times R_{it}$.

3.1.3 Data on democracy.

Regime type is measured through the Polyarchy/Electoral Democracy index, corresponding to `v2x_polyarchy`, provided by the Varieties of Democracy (V-Dem) dataset ([V-Dem Institute, 2024](#); [Coppedge et al., 2024](#)). This measure ranges from 0 to 1, with higher values representing greater democracy. Annual observations covering almost all sovereign states since the mid-20th century are available in the V-Dem database. In the current panel, the index has no missing values in any of the 10,175 country-year observations.

3.1.4 Data on economic and demographic control variables.

Economic and demographic control variables are taken from the World Bank’s World Development Indicators (WDI) database ([World Bank, 2024](#)). These include GDP in constant U.S. dollars, GDP_{it} , corresponding to `gdp_kd` (WDI series NY.GDP.MKTP.KD); population, Pop_{it} , corresponding to `pop_total` (SP.POP.TOTL); military expenditure, corresponding to `mil_exp_cd` (MS.MIL.XPND.CD, from SIPRI); military expenditure as a share of GDP, M_{it} , corresponding to `mil_exp_gdp_pct` (MS.MIL.XPND.GD.ZS); exports as a share of GDP, corresponding to `exports_gdp_pct` (NE.EXP.GNFS.ZS); and industry value added as a share of GDP, I_{it} , corresponding to `industry_gdp_pct` (NV.IND.TOTL.ZS).

GDP and population indicators have 91.2% and 98.8% country-year coverage in the panel. The coverage is less complete for other variables, including those measuring the composition of the economy. This can be explained by the fact that there is a lot of missing data for small states and conflict zones for the earlier decades.

3.1.5 Panel structure.

The total number of observations from the composite panel is 10,175 for the years ranging from 1963 to 2024 in 176 countries. Table 1 shows the summary statistics for the primary variables. The panel is an unbalanced one since some new states have been added as well as due to differences in data availability, but the key variables have full coverage in the sample.

Table 1: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Annual GEO entries, n_{it}^{GEO}	10,175	0.090	0.712	0	18
Any GEO entry, A_{it}^{GEO}	10,175	0.032	0.177	0	1
Rival GEO share, R_{it}	10,175	0.014	0.087	0	1.000
Active rival count, C_{it}	10,175	0.736	1.250	0	8
V-Dem polyarchy	10,175	0.426	0.286	0	0.923
$\ln(\text{GDP}_{it})$	9,283	24.07	2.181	18.15	30.75
$\ln(\text{Pop}_{it})$	10,056	15.77	1.765	10.72	21.10
Military expenditure/GDP, M_{it}	7,540	2.66	3.21	0	117.4
Industry/GDP, I_{it}	7,566	27.53	11.86	0	86.67

Notes: Panel covers 176 countries, 1963–2024. GDP is in constant U.S. dollars; military expenditure shares and industry shares are percentages of GDP. The rival GEO share R_{it} ranges from 0 (no rival GEO presence) to 1.0 (rivals own all recent global GEO payloads).

3.2 Baseline OLS specification

I begin with a descriptive panel model with year fixed effects:

$$n_{it}^{\text{GEO}} = \beta R_{it} + X_{it}\Gamma + \lambda_t + \varepsilon_{it}, \quad (1)$$

i refers to countries, and t to years. The dependent variable is defined as the number of GEO entries per year. The independent variable is defined as the rival share indicator, calculated using a five-year moving average. Controls used in the model include the log of GDP, log of population, and military spending. Standard errors are clustered by country. The current model is merely descriptive and not causal; its purpose is solely that of illustrating the relationship within the panel data before the design component of the paper.

3.3 Main design: staggered rival-pressure jump DID

The main empirical approach used in this research paper is the staggered difference-in-difference approach, where there is a test to see if a sudden increase in rival pressure leads to a change in the intensive margin of GEO behavior. To do this, the continuous variable for rival pressure is converted into a treatment time variable.

Rival pressure, expressed in percentage points, is defined as:

$$R_{it}^{\text{pct}} = 100 \times R_{it}. \quad (2)$$

To capture sudden shifts in the strategic environment rather than gradual drift, I define a treatment shock as the difference between the current level of rival pressure and its average over the preceding three years:

$$J_{it} = R_{it}^{\text{pct}} - \frac{1}{3} \sum_{s=1}^3 R_{i,t-s}^{\text{pct}}. \quad (3)$$

The treatment year, T_i , is defined as the first year in which a country experiences a substantive shock, operationalized as $J_{it} \geq 0.2$ percentage points. The estimation sample is restricted to the 1970–2020 window and retains treated units with sufficient support for dynamic analysis, requiring at least five years of pre-treatment and five years of post-treatment observations. Under this restriction, the DID sample includes 29 treated countries and 143 never-treated controls.

The dependent variable is the intensive margin of GEO activity, measured as

$$y_{it} = \log(1 + n_{it}^{\text{GEO}}), \quad (4)$$

where n_{it}^{GEO} is the annual number of GEO entries attributed to country i in year t . This transformation reduces the influence of extreme counts while preserving the distinction between zero-entry and positive-entry country-years.

The baseline specification is a two-way fixed effects DID model of the form

$$y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}, \quad (5)$$

with $D_{it} = \mathbb{1}[t \geq T_i]$ if the country is treated, and zero otherwise; α_i represent country fixed effects, and λ_t are year fixed effects. The standard errors are clustered by the country. The identification strategy then becomes changes within countries following a shock to rival pressure, compared with how non-treated or never-treated counterparts evolve over the same period. In addition to the standard TWFE regression analysis, I apply an event study design based on relative event times, allowing for a dynamic impact after treatment and checking for differences in pre-treatment trends in the periods just before treatment.

It has been acknowledged that standard TWFE estimators can potentially be flawed when dealing with a staggered design owing to the possibility of the presence of heterogeneity across different cohorts and/or through time. In particular, [Goodman-Bacon \(2021\)](#) claim that the TWFE estimator consists of averages of several underlying 2×2 DID estimators, while [de Chaisemartin and D’Haultfœuille \(2020\)](#) and [Sun and Abraham \(2021\)](#) show that some of these estimators might suffer from contamination due to having treated individuals serve as control groups for future treated individuals. Therefore, the baseline TWFE estimates get augmented with an interaction-weighted correction introduced by [Sun and Abraham \(2021\)](#). The procedure followed by the Sun–Abraham estimator filters out comparisons made with control group observations which have not been nor are treated at the point of comparison to prevent any contamination arising from the forbidden comparison rule. In this particular application, the SA estimator represents both a robustness check on the main DID specification and an alternative benchmark for the estimation of dynamic

treatment effects. The decomposition by Goodman-Bacon is described in the appendix C.

4 Baseline OLS and Main DID Results

4.1 Baseline OLS Results

Table 2 represents a baseline OLS regression analysis including yearly fixed effects. In terms of our dependent variable, we should mention n_{it}^{GEO} , which denotes the number of GEOs per year. As far as our main predictor (R_{it}) is concerned, its coefficient is both statistically significant and positive. Interpreting this finding, we may argue that if our measure of rivals' share increases by 1 unit, then we should observe 1.281 more annual entries per year in each given country, while clustering error estimate is 0.584. Considering the meaning of the dependent variable in our regression model, we may claim that the described coefficient is rather high and understandable.

Speaking about other independent variables, although there are some small differences, there is no doubt about their importance. First of all, let us say that log-population coefficient is highly significant and positive, which means that bigger countries have greater numbers of entries annually. At the same time, military expenditures are positively related to the dependent variable, and this finding fits the logic well since such actions require certain state capabilities and resources. Log-GDP coefficient, in turn, turns out to be insignificant and very close to zero, which means that GDP itself cannot explain GEO entry activities when other factors are controlled.

All of the presented estimates should be interpreted in a purely descriptive manner. It should be noted that I deliberately excluded country fixed effects from Table ??, because this analysis is performed in order to illustrate a broader correlation between rival pressure and GEO entries. Adding country fixed effects, in this case, would make our estimands to focus on the within-country change in the outcome, which corresponds exactly to the DID framework below.

The effect of the rival share variable becomes smaller as controls are added, dropping from 4.277 in column (1) to 1.281 in column (4); however, the coefficient still retains its positive significance in the model with the highest number of controls. While some of the raw correlation can be explained by the varying sizes and resource stocks of the countries involved, not all of it can be explained in this way. Thus, the rivalry variable seems to reflect something beyond a mere stand-in for scale. The increase in R^2 is small between columns (1) and (3), but substantial in column (4) with the addition of military spending—as the wider discussion would suggest, GEO behavior is driven not only by rivalry but by state capability.

4.2 Main DID Result: TWFE and Sun–Abraham

I now move on to discuss the design of the paper. The key empirical question is whether a discrete rise in the level of rival pressure leads to a corresponding discrete change in the intensive margin of GEO participation. The staggered DID setup involves contrasting the evolution of log annual GEO entries of countries that have undergone a discrete rise in rival pressure to the same trend among

Table 2: Baseline OLS Results: Incremental Specifications

	Dependent variable: Annual GEO entries			
	(1)	(2)	(3)	(4)
Rival share (5-year window)	4.277*** (1.602)	3.668*** (1.342)	3.604*** (1.344)	1.281** (0.584)
Log GDP		0.0872* (0.0469)	0.0714 (0.0457)	-0.0030 (0.0138)
Log population			0.0310** (0.0133)	0.0288** (0.0111)
Military expenditure (bn USD)				0.0132*** (0.000567)
Year fixed effects	Yes	Yes	Yes	Yes
Observations	7,570	7,570	7,570	7,570
R^2	0.174	0.220	0.222	0.639

Notes: This table reports baseline OLS estimates of the association between rival pressure and annual GEO entry counts. The dependent variable is the annual number of GEO entries recorded by country i in year t . All specifications include year fixed effects, and standard errors clustered at the country level are reported in parentheses. Columns (1)–(4) add controls sequentially: log GDP, log population, and military expenditure. Military expenditure is measured in billions of current U.S. dollars. The estimation sample is held constant across specifications for comparability. These estimates are descriptive and should not be interpreted causally.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

countries without such a rise, controlling for country and year fixed effects. While the estimand is not related to rivalry’s role in explaining entry into GEO, it is focused on whether countries subject to a discrete strategic shock subsequently increase their GEO activities compared to similar counterfactuals. This interpretation is purposely limited because it is consistent with Hypothesis H1 in Section 2.3, predicting a modest intensive margin effect in already capable countries.

Table 3 shows the static ATT from both the baseline TWFE estimator in Equation (5) and the interaction-weighted estimator of Sun and Abraham (2021). The sample used in the analysis comprises 8,211 country-year observations between 1970 and 2020, including 29 treated countries and about 143 never-treated countries.

Table 3: Main DID Result: TWFE vs. Sun–Abraham (IW)

Estimator	Coefficient ($\hat{\beta}$)	Std. Error	t -value	p -value
TWFE (baseline)	0.0611	0.0261	2.339	0.0205
Sun–Abraham (IW)	0.0614	0.0360	1.708	0.0894

Notes: The dependent variable is $\log(1 + n_{it}^{\text{GEO}})$. Both specifications include country and year fixed effects. Standard errors are clustered at the country level. The SA estimate is aggregated across all post-treatment periods using the IW weights from Sun and Abraham (2021). The estimation sample is the 1970–2020 panel with 29 treated units and 143 never-treated controls.

The baseline TWFE specification yields

$$\hat{\beta}_{\text{TWFE}} = 0.0611 \quad (SE = 0.0261, p = 0.0205). \quad (6)$$

Since the dependent variable is $\log(1 + n_{it}^{\text{GEO}})$, it should be interpreted in terms of proportion rather than levels. An estimate implies an increase of $e^{0.0611} - 1 \approx 6.3\%$. Hence, a rivalry shock results in a 6.3 percent increase in GEO satellite program development—a relatively minor effect. The size of the estimated coefficient can be viewed as another reason for its credibility. Satellite programs take several years to develop and require substantial upfront spending. A relatively moderate reaction to a rivalry shock is much more believable than a highly exuberant one. This observation resonates with the mechanism 3 presented in Section 2.3: rivalry acts as an accelerator of already-existing capabilities.

First, notice the extremely close correspondence between the TWFE and SA estimates. Their discrepancy of less than 0.5% suggests that in this particular data set the negative weighting bias does not substantially distort the average difference-in-differences. Namely, even though some of the comparisons may have very high weights, their number is sufficiently small. The GB decomposition of the weights presented in Appendix C confirms this hypothesis. Second, since forbidden comparisons are discarded from the analysis, it is natural that the Sun–Abraham estimate’s standard error is larger than the TWFE standard error. Specifically, while the TWFE SE is 0.0261, the SA SE equals 0.0360. Correspondingly, the p -value for the SA coefficient is 0.0894, whereas that for the TWFE is 0.0205.

4.3 Dynamics of the Event Study

It is important to understand that although the static DID coefficient is useful in estimating the average impact of the rival-pressure shock, it is insufficient to convey the entire information content of the difference-in-differences estimate. The event study regression can help us better understand both the dynamics of the response as well as detect possible differential pre-treatment trends.

Figure 1 presents dynamic coefficients estimated by the Sun–Abraham estimator over a window of 10 years before and after the treatment year, $[-5, +5]$. The coefficients corresponding to the pre-treatment period ($t = -5$ through $t = -2$, relative to the omitted treatment period of $t = -1$) are essentially zero, which is illustrated by a nearly horizontal trend line. It indicates that there are no significant differences between the dynamics of control and treated groups prior to receiving the rival pressure shock. More formally, we test for joint equality of all pre-treatment SA coefficients, $\{-5, -4, -3, -2\}$, to see if there are any differential pre-treatment trends in the data. The Wald test yields $\chi^2(4) = 0.663$ with $p = 0.999$.

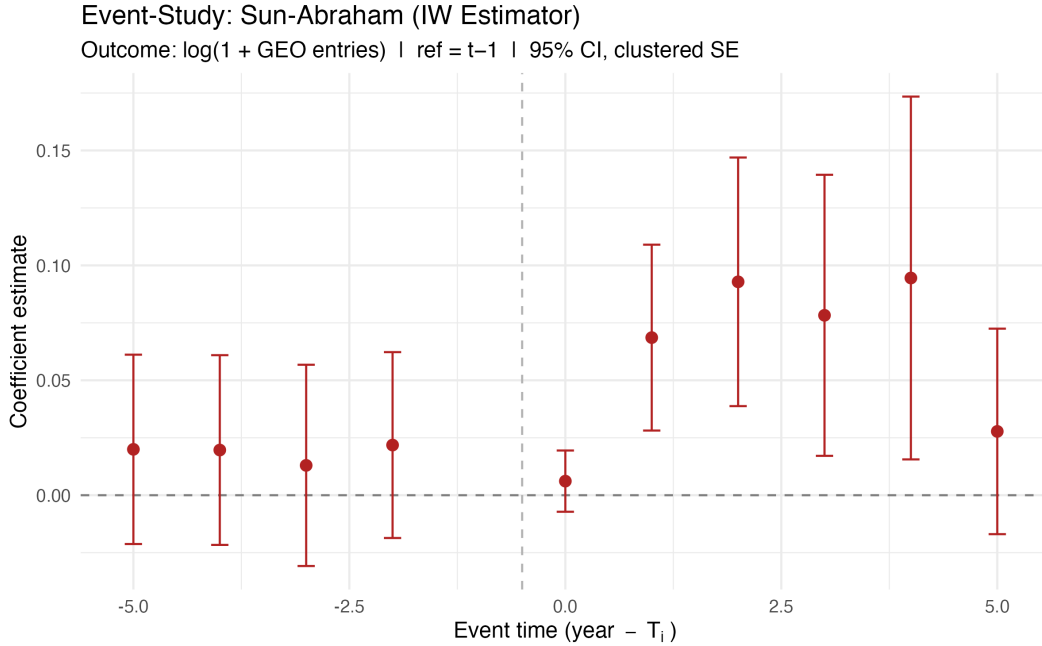


Figure 1: Dynamic event-study using the Sun–Abraham (IW) estimator. Coefficients are estimated relative to $t = -1$. The dependent variable is $\log(1 + n_{it}^{\text{GEO}})$. Error bars represent 95% confidence intervals with country-clustered standard errors.

The post-treatment response profile generated by the SA estimator is a delayed yet positive profile. The effect of the treatment at $t = 0$ is relatively weak, but it grows progressively over $t = +2$, $+3$, and persists up to $t = +4$. Such a pattern is precisely what should be observed in the case of space policy responses to rivalry because there are multi-year processes of satellite procurement, production, and launches. Therefore, the manifestation of the materialized response of states will take some time and appear in the actual numbers of GEO entries as predicted by

the theory of Section 2.3 (mechanism 3). For the graphical comparison of the two estimators, see Appendix B and the TWFE event study plot; as expected, the SA plot demonstrates a much cleaner profile, especially before the treatment, which supports the point raised in Sun and Abraham (2021) in regard to the problem with comparisons of the fixed effects model estimates.

4.4 Magnitude Interpretation

Careful interpretation of the effect estimate is necessary due to the peculiar nature of the outcome variable. The variable $\log(1 + n_{it}^{\text{GEO}})$ condenses count information and, in essence, becomes linear around zero. In fact, the majority of country-year observations have $n_{it}^{\text{GEO}} = 0$; the average number is just 0.09, while the proportion of observations with non-zero GEO launches amounts to just 3.2%. Under such conditions, an estimate of 0.061 in logs implies an increase in the outcome variable by 6.3%; although numerically small, this is a relatively large percentage change for countries that launch their satellites into GEO orbits.

Still, the effect has to be interpreted conditionally. According to Section 5.3, the impact of rival pressure on satellite launches appears to be significantly larger ($\hat{\beta} \approx 0.20$) among countries that conduct space missions. Therefore, when a state launches one or two satellites annually, the estimated 0.20 log effect translates to a 22% increase per year in the number of satellite launches after the rival pressure shock (given $\exp(0.20) - 1$). It is an interesting finding that suggests the effect of rivalry on satellite procurement policies even after accounting for country- and year-fixed effects. Furthermore, the effect is rather plausible since it is conditional on the actual increase in the ratio of rival pressure—that is, 0.2%.

5 Sensitivity and Robustness Analysis

The current subsection studies the robustness of the main DID estimate against variations across the following aspects: pre-trends test, placebo test via permutations, treatment effects heterogeneity based on pre-treatment capacity, leave-one-cohort-out sensitivity analysis, sensitivity analysis based on a treatment cutoff point choice, and TWFE vs. event study. In summary, the purpose of this exercise is to assess the extent to which the result obtained in Section 4, where the authors argue that the rivals produce an uptick in shocks followed by gains in GEO entry intensity, remains unchanged in face of changes in model specifications.

5.1 Pre-Trends Diagnostics

Difference-in-Differences requires only one assumption — the *parallel trends* assumption. Namely, in absence of a treatment effect, both control and treatment groups should demonstrate the same dynamics of their outcomes. Traditionally, the check for parallel trends consists of performing a joint Wald test of whether the set of pre-event leads $\{\alpha_1, \alpha_2, \dots, \alpha_T\}$ is simultaneously equal to zero.

Test of the null hypothesis $\beta_{-5} = \beta_{-4} = \beta_{-3} = \beta_{-2} = 0$ based on the standard Wald framework

gives the following results:

$$H_0 : \beta_{-5} = \beta_{-4} = \beta_{-3} = \beta_{-2} = 0, \quad \chi^2(4) = 0.220, \quad p = 0.928.$$

On the other hand, the SA-based test described in Section 4 yields the following χ^2 statistics:

$$\chi^2(4) = 0.663, \quad p = 0.999.$$

Both tests fail to reject the null that the pre-treatment coefficients are jointly equal to zero. This does not prove parallel trends, but it provides no evidence of differential pre-treatment movement within the observed pre-window. This result makes the conclusion obtained in Section 4 more credible since the SA test uses only clean control samples.

However, three important caveats must be mentioned. First, the test based on the Wald framework lacks statistical power because there are just 29 treated observations (across various cohorts). Second, only a five-year horizon before the shock was considered. Violations of parallel trends at other horizons could go undetected by such a test. Third, standard TWFE estimators suffer from contamination problems described in [Goodman-Bacon \(2021\)](#) and [Sun and Abraham \(2021\)](#). For this reason, sensitivity analysis performed in Section 4 is based on SA-based event-study coefficients.

5.2 Placebo Test

In order to check whether the result obtained can be related to some kind of non-causal factors rather than the true causal effect of treatment on the outcome, the authors perform a placebo test through random permutations. In particular, in each of 1,000 repetitions, treatment status was randomly assigned to a randomly chosen subset of 29 countries from the full sample while keeping constant distribution of cohort years. Then, for each repetition, the core TWFE DID equation was estimated and a placebo coefficient on the rival-pressure shock term was computed.

The placebo distribution reported in Appendix Figure 8 is slightly shifted from zero (its mean equals 0.005) and is approximately symmetric, as expected if treatment has no effect. The actual estimate is slightly positive ($\hat{\beta} = 0.0611$) and lies in the right tail of the placebo distribution. Specifically, in 16 out of 1,000 repetitions, coefficients on the rival pressure shock are larger than the observed coefficient. Hence, a permutation p-value is 0.016. Notably, the advantage of a permutation test is that it complements the standard regression inference, where clustering is used to account for dependence among observations within countries. However, in a case when only 29 countries were included, asymptotic approximation becomes problematic.

5.3 Treatment Effect Heterogeneity by Capacity

The theoretical model outlined in Section 2.3 suggests that rivalry matters relatively more to those states already endowed with institutional and industrial capacity to respond to strategic pressures. This is hypothesis H2. The aggregate DID coefficient averages across both treated and never-

treated countries irrespective of their capacity levels. To evaluate the capacity-based hypothesis, I divide the treated group into four subgroups: (i) high-GDP countries that had a high median of $\ln(\text{GDP})$ during the five years preceding treatment, (ii) low-GDP countries, (iii) countries that previously launched satellites, and (iv) those that did not.

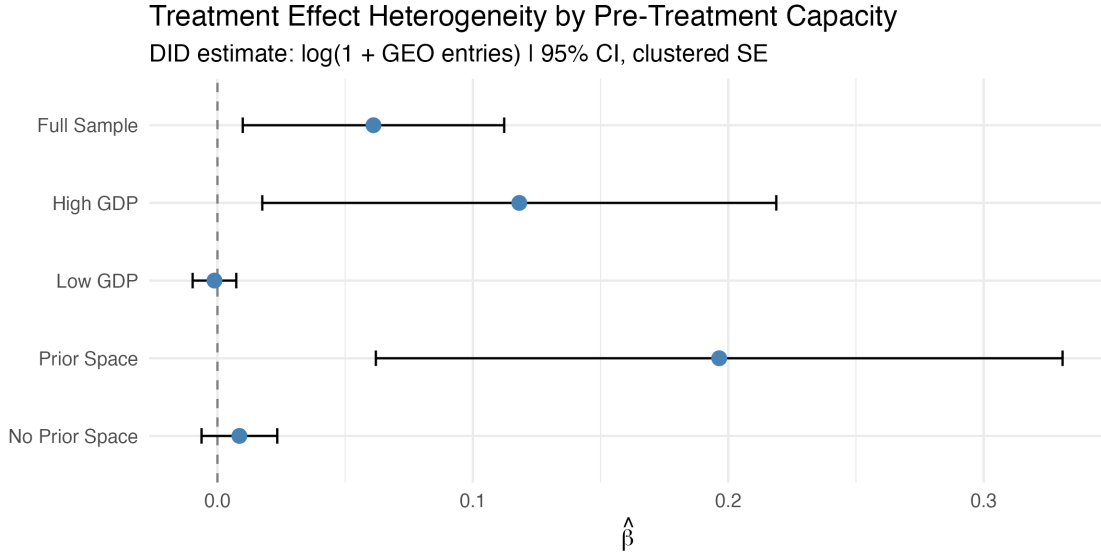


Figure 2: Treatment effect heterogeneity by pre-treatment capacity. Each row shows the DID coefficient obtained in a separate regression constrained to the corresponding subgroup of treated countries, along with controls for never-treated countries. Error bars correspond to 95% confidence intervals using country-clustered standard errors.

Figure 2 plots the results. For high-GDP treated units, the point estimate is about $\hat{\beta} \approx 0.12$, nearly twice the estimate for the overall sample. In contrast, the estimate for the low-GDP group of countries is close to zero ($\hat{\beta} \approx 0.00$), suggesting that economically weaker countries are insensitive to rivalry shocks. The heterogeneity is even more pronounced for subgroups divided on the basis of pre-treatment space activities: whereas the presence of prior space experience implies a significant effect ($\hat{\beta} \approx 0.20$), the absence thereof means an essentially zero effect. In short, the heterogeneity analysis fully vindicates H2: the observed effect of 0.06 is not the average response across treated units but an aggregation of effects concentrated only among states with pre-treatment capacity for launching satellites.

Another confounding variable addressed through heterogeneity analysis concerns how the rival-pressure jump may be driven by country importance unrelated to rivalry. One could argue that the rise in rival pressures just indicates that a country was “important” and therefore had a tendency to expand GEO activities for reasons other than strategic rivalries. In this case, the effect would manifest itself across both subgroups with and without pre-treatment capacity. That it does not happen strengthens the conclusion about rivalry driving the empirical finding rather than country importance per se.

5.4 Leave-One-Cohort-Out Sensitivity

It is possible that the aggregate effect can be attributed to the idiosyncrasies of one particular treatment cohort. To test this idea, I re-run the TWFE specification 15 times, each time omitting all treatment units from a single cohort. The findings are reported in Appendix Figure 9. The coefficient stays consistently positive in all specifications, varying within a relatively narrow range (between approximately 0.045 and 0.070). No exclusion of one particular cohort brings the estimate to either zero or makes it statistically insignificant and/or turns it into a negative number. The greatest variation is associated with exclusion of the 1996 cohort ($n = 5$, which is the largest one) or the 1984 cohort, since omission of more treated observations mechanically pushes the estimate down. Therefore, the finding is not contingent on a specific group of countries.

5.5 Treatment Threshold Sensitivity

The specification in Section 4 defines treatment as occurring at the first point in time when rival pressure increases by at least 0.2 percentage points relative to the previous period. Of course, the threshold is a researcher choice and the effect should not be sensitive to the exact value chosen. I re-estimate the TWFE and SA models at different levels of the Δ threshold: $\Delta \in \{0.10, 0.15, 0.20, 0.25, 0.30\}$. The results are presented in Appendix Figure 10. Consistent with the theory, the coefficients stay positive across all considered thresholds, tracking each other very closely. They increase slightly with Δ : the estimate goes from approximately 0.04–0.05 at $\Delta = 0.10$ to 0.07–0.08 at $\Delta = 0.30$. Confidence intervals also widen with increasing Δ due to shrinking treated groups, yet the conclusion does not change. Importantly, TWFE and SA estimates never diverge for any of the considered thresholds, implying that the TWFE estimator is not biased toward negative weighting in this setting.

Summary. Along all six robustness dimensions considered—pre-trends diagnostics, placebo permutation, capacity heterogeneity, leave-one-cohort-out sensitivity, treatment threshold variation, and TWFE-SA comparison presented above—the core finding withstands scrutiny. The effect of rival-pressure shocks on the intensity of entry into the GEO region is not an artifact of negative weighting in the TWFE estimator, not an idiosyncrasy of any particular treatment cohort, not sensitive to the threshold used to define treatment, and is concentrated among the subgroup of capacity-endowed states as predicted by H2.

6 Conclusion

While there is much value in the estimates reported above, the principal contribution of this paper is the careful consideration of when such rivalry-induced intensification can be expected to take place. On the intensive margin of GEO entry, the presence of rivalry increases the frequency of GEO entries by about 6% in the full sample and by 20% among previously-active countries, respectively. The Sun–Abraham correction establishes that this relationship is not due to negative weighting

in the TWFE estimator, the placebo permutation test eliminates chance ($p = 0.016$), and the leave-one-cohort-out analysis confirms that the effect is not driven by a particular set of treatment cohorts. Critically, the heterogeneity analysis reveals that the overall effect is the weighted sum of a strong response among already-capable states and a virtually nonexistent response in other cases. Appendix E further demonstrates that on the extensive margin of first-time GEO entry, rivalry loses importance after accounting for basic capability indicators, as hypothesized in H3.

Taken together, these findings suggest a reassessment of what it means for states to compete in space. Policy discourse around the subject often takes rivalry for granted, assuming a zero-sum game in which a state that sees a rival’s orbiting assets will feel compelled to respond with similar capabilities. These data challenge that assumption. Rivalry is not capability: states can recognize the significance of a rival’s capabilities and yet lack the means to respond to it. Entering GEO requires both a high level of space capability and a considerable budget to finance the process, as well as organizational coordination to ensure a successful mission. Hence, it seems logical that while basic capability indicators play the most important role on the extensive margin—because they determine whether a country has the option to participate in GEO in the first place—rivalry exerts influence on the intensive margin—because the question is whether the state wants to enter faster. Such a conditional-accelerant interpretation is consistent with the discussion in Section 2.3 and with the general space-security literature (Schrogl et al., 2015; Pekkanen and Blount, 2024; Moltz, 2019).

These findings also serve to nuance the metaphor of arms racing that dominates both scholarly and popular writing about space competition. To be sure, there is something to it: the DID approach provides evidence that some form of escalatory behavior takes place among active states in space. Nevertheless, the metaphor becomes problematic when it fails to acknowledge feasibility constraints. While a country equipped with launch capabilities, engineering expertise, and budget surplus is able to respond to rival pressure shocks through accelerated missions, a country lacking the required infrastructure cannot make a decision and enter into GEO on a whim.

Several limitations warrant consideration. First, while the threshold sensitivity analysis establishes that the results are relatively robust across various treatment definitions, the treatment is nevertheless constructed by the researcher rather than occurring naturally. Second, since only 29 treatment groups are included in the main DID regression, statistical power might be an issue. SA estimates feature noticeably wider standard errors compared to TWFE counterparts and reach the point of significance only at the 10% level. Third, the measure of GEO activities considered in this paper—the number of entries—is a coarse indicator of space capabilities. It does not discriminate between small communication satellites and large military surveillance platforms and fails to reflect investments in infrastructure, spectrum allocation, and launch vehicle development. Finally, the rivalry measure adopted here is calculated based on GEO assets held by strategic rivals, which themselves emerge from the process in question.

All these caveats notwithstanding, this paper provides solid empirical insight. Namely, using the DID and SA approaches, it presents design-based evidence consistent with a modest causal relationship between geopolitical rivalry and the intensive margin of GEO activity. The effect

appears to be highly sensitive to states' capabilities. An understanding of this key feature of space competition is critical when evaluating the existing accounts. Future research might extend the framework presented here by exploring other orbits (such as LEO or MEO) or using more refined capability measures.

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A Treatment Cohort Distribution and Pre-Normalization Event Time Means

Figure 3 depicts the treatment cohorts distribution in our 1970-2020 period of analysis. Out of 29 treated units meeting the support assumptions, the most numerous treatment cohort takes treatment around the mid-1990s, namely in 1996 when there are already five countries with their first rival pressure increase. The second peak appears in the late 2000s/early 2010s, when new space-capable states emerge and rivalries become active in Asia and the Middle East region. Treatment staggered over four decades ensures that the TWFE estimate is not affected by any particular historical period and sufficient SA estimator’s cross-cohort variability in time is achieved.

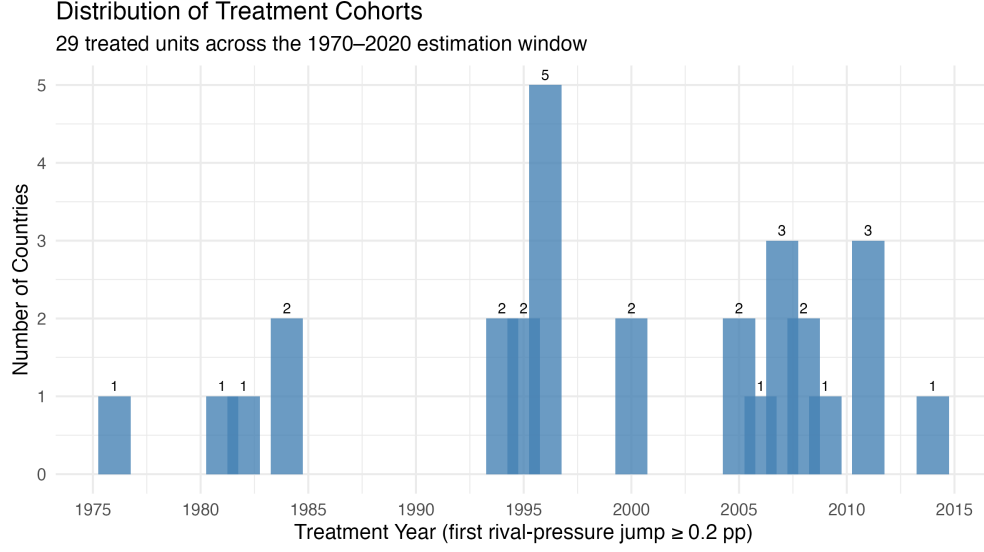


Figure 3: Treatment cohorts distribution. Number of countries experiencing their first rival pressure jump (≥ 0.2 percentage points) in each year. 29 treated units start treatment ranging from 1976 to 2014.

Figure 4 illustrates the pre-normalization event time means of the treated cohorts. This is measured as demeaned outcome series relative to pre-treatment average (i.e., $t < 0$). As expected, in the pre-treatment window, the series is close to zero, meaning that no systematic trend in the GEO activity is observed prior to the rival pressure shock. Once the treatment is received, there is a positive increasing trend until the 2nd-3rd event-time.

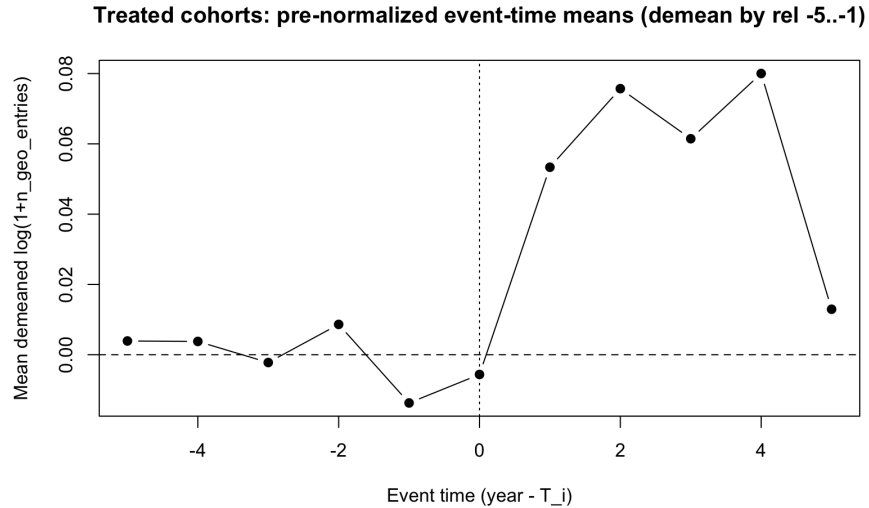


Figure 4: Pre-normalization event time means for treatment cohorts. Outcome is $\log(1 + n_{it}^{\text{GEO}})$ demeaned by pre-treatment average (periods -5 to -1). The vertical dashed line indicates the treatment year.

B TWFE Event Study and Comparison to Sun–Abraham Approach

Figure 5 presents the standard TWFE event study plot. This graph estimates coefficients for each event time relative to $t = -1$. Coefficients for pre-treatment event times cluster around zero while post-treatment event times exhibit much higher variance compared to those estimated by the SA approach in the main text. Figure 6 compares both event studies in the same graph. As can be seen, at long event times, the TWFE profile starts drifting downwards and crosses zero line while the SA profile exhibits an increasing trend up to the 4th event time. This behavior corresponds to what is discussed by [Sun and Abraham \(2021\)](#) regarding contamination of the TWFE estimate by forbidden comparisons (whereby the treated before serves as implicit control to treated after). This implies that the SA profile represents the true dynamic effect of the treatment.

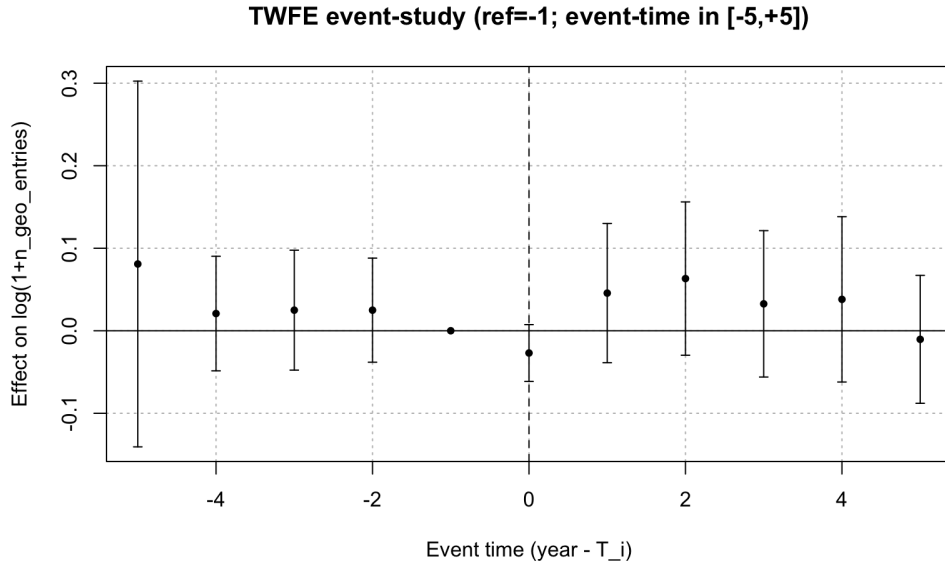


Figure 5: TWFE event study for rival pressure. Event time estimates of coefficients relative to $t = -1$. Dependent variable is $\log(1 + n_{it}^{\text{GEO}})$. Error bars are 95% confidence intervals.

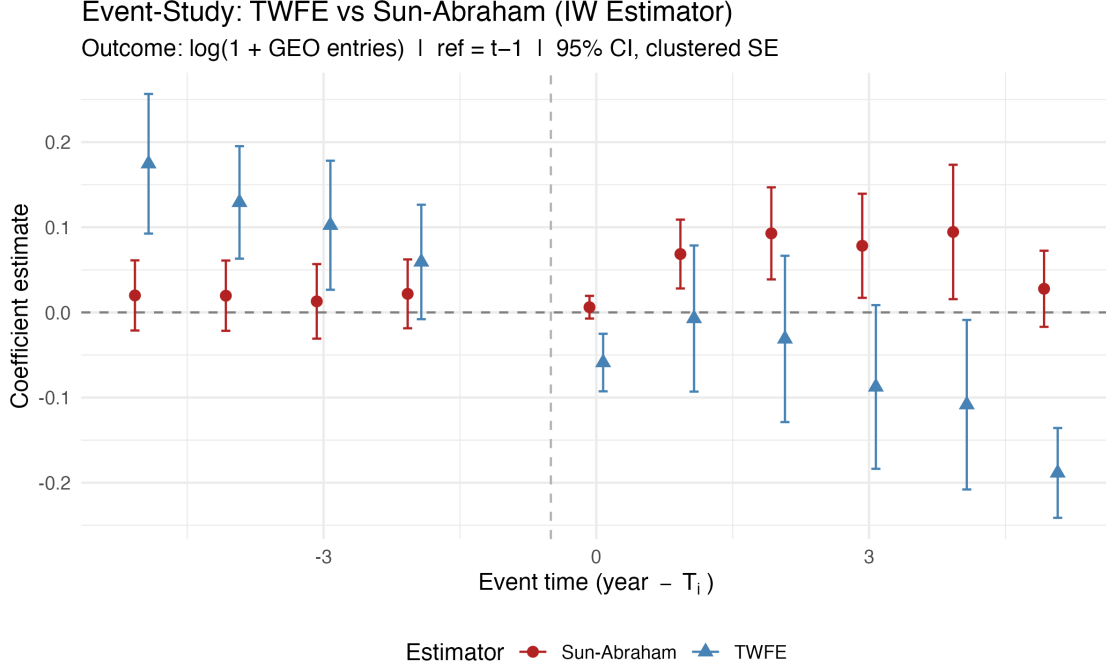


Figure 6: TWFE versus Sun-Abraham event studies. TWFE event profile starts to drift downwards at event times beyond $t = +2$, and eventually becomes negative (as it includes forbidden comparisons). SA coefficient is monotonically increasing up to $t = +4$.

C Goodman–Bacon Weighted DID Comparisons Decomposition

Figure 7 presents the Goodman-Bacon (2021) decomposition of TWFE estimate into weighted sum of 2×2 DID estimators, whereby dots mark 2×2 DID comparisons, y-axis is estimate, x-axis – weight. Most of TWFE estimator weight is associated with the comparisons between treated and untreated groups (green dots) that fall around TWFE estimate, which is close to zero. Comparisons between later versus earlier treated groups (red triangles) and earlier versus later treated groups (blue dots) have smaller weight and located far away from TWFE center line. This means that the extreme comparisons are weighted negligibly small and therefore they do not affect the estimate. This result is consistent with near equivalence of TWFE and SA estimates demonstrated in the main text.

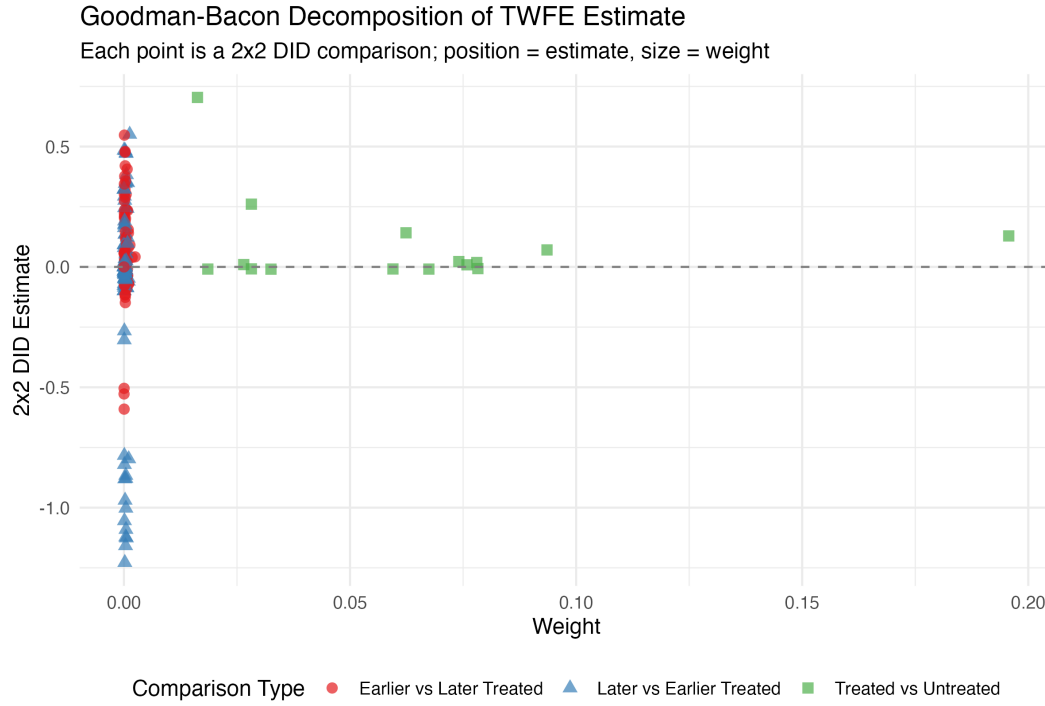


Figure 7: Goodman-Bacon decomposition of TWFE estimate. 2x2 DID comparison dots mark each estimate on y-axis, its weight on x-axis. Estimates dominated by treated versus untreated comparisons (green dots) concentrated around TWFE coefficient.

D Placebo Test, Leave-One-Cohort-Out Sensitivity and Threshold Robustness Analysis

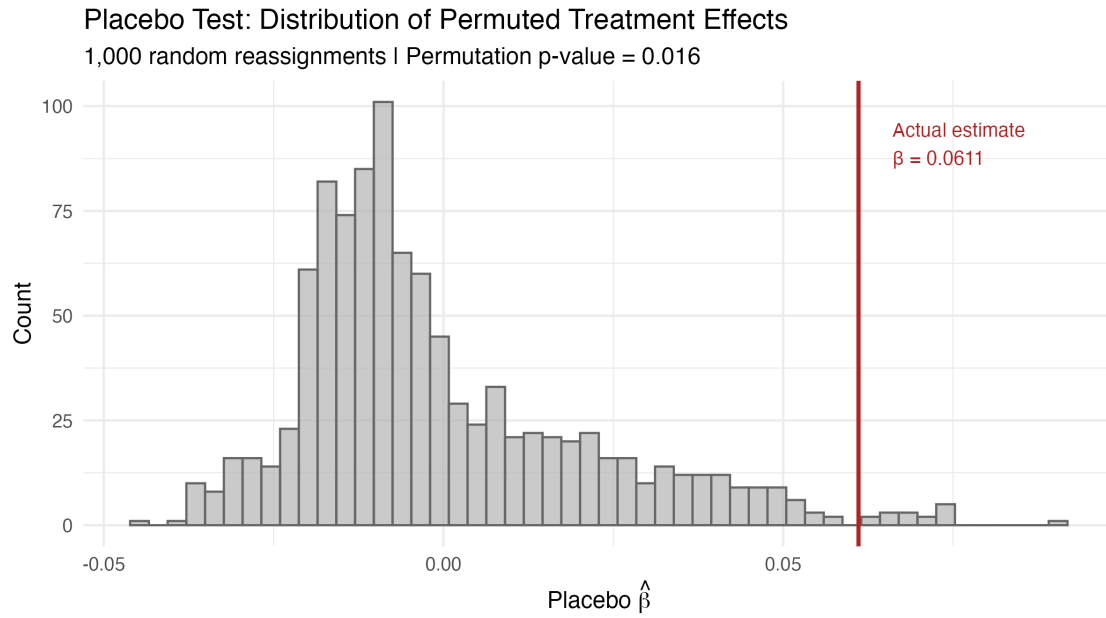


Figure 8: Random permutation placebo test for the DID estimate. Histogram is based on 1,000 random draws from a distribution of coefficient drawn via permutations of rivals-share vector. Red vertical line indicates the real DID coefficient $\hat{\beta} = 0.0611$. Random permutation p -value = 0.016.

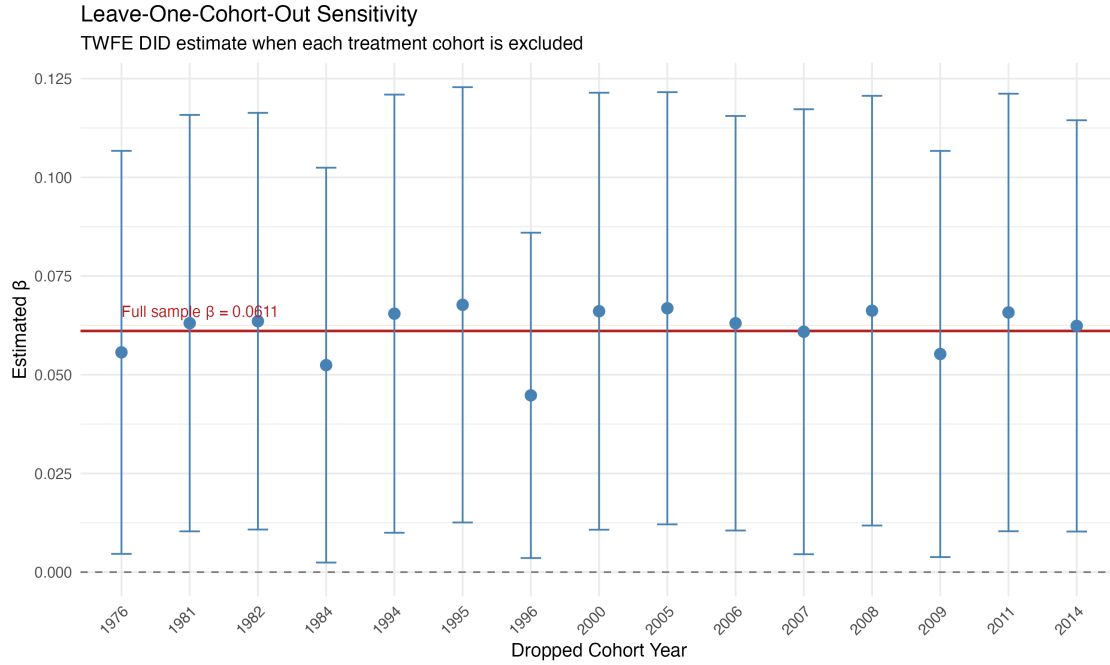


Figure 9: Leave-one-cohort-out sensitivity for TWFE estimate. TWFE DID coefficient estimate without the selected treatment cohort (dots). The red line is TWFE estimate based on full sample ($\hat{\beta} = 0.0611$). Country clustering standard errors and error bars for 95% CI.

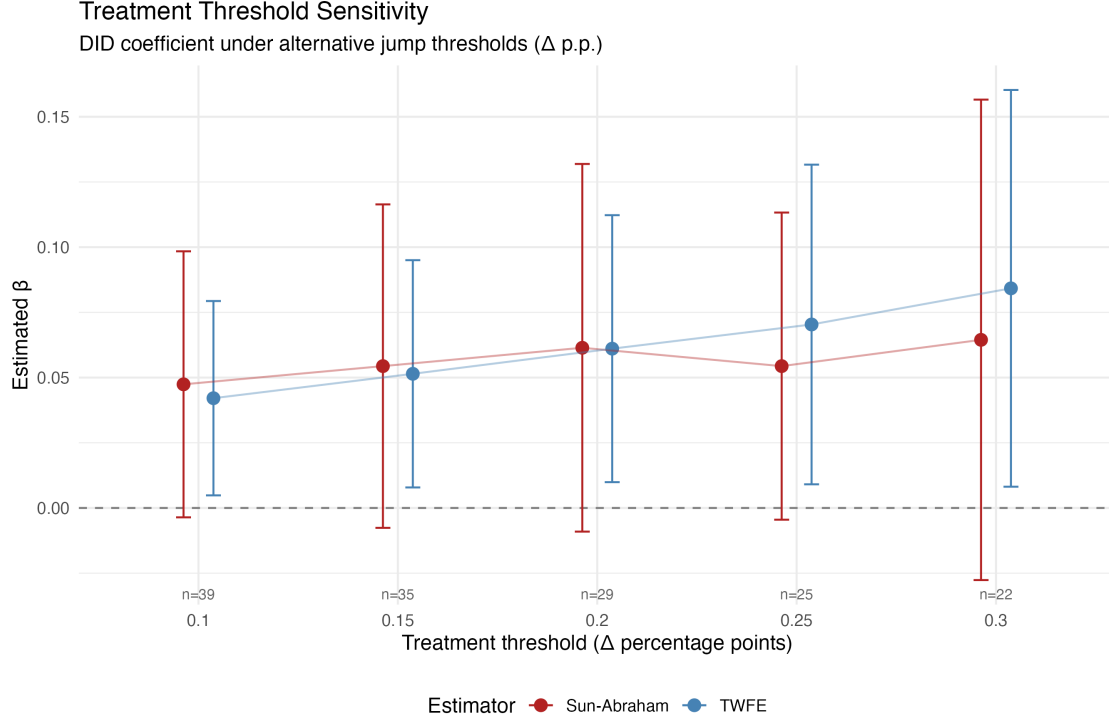


Figure 10: Treatment threshold sensitivity analysis. TWFE and SA coefficients’ estimates (point pairs) at the selected Δ threshold. The number of treated units decreases as the threshold increases: from $n = 39$ at $\Delta = 0.10$ to $n = 22$ at $\Delta = 0.30$. Error bars represent 95% CI bounds.

E Supplementary Cox Survival Model of the First GEO Entry

In this section, we carry out Cox survival analysis replication of our findings in DID specification on the intensive margin in the framework of H3 presented in the main text. We demonstrate that the rivalry variable does not explain GEO activity on the extensive margin, once controlling for the capacity measures. We further elaborate the theoretical considerations provided in Section 2.3 on the lack of an effect.

In order to analyze H3 of Section 2.3, we conduct standard Cox proportional hazards analysis for first-ever GEO deployment in 176 countries with 9,278 observations and 36 failures. Country-level clustered standard errors are used.

From Significant Hazard Ratio to Non-significant: Effect of Controlling for Capabilities

Figure 11 shows Kaplan-Meier survival curves in relation to rivalry exposure. Specifically, we separate the units by maximum values of R_{it}^{pct} experienced over their observed history. We consider the unit to be rivalry exposed (has been exposed to at least one instance of rivalry over time period) if its max rivalry exposure is above zero (“yes”: 23 countries), otherwise “no rivalry exposure” (“no”:

130 countries).

In this uncontrolled specification, the distinction is quite sharp: countries with high rivalry exposure enter GEO twice as fast as the others (log-rank test $p = 0.0068$). However, the association of rivalry exposure with first GEO entry is confounded by countries' basic capabilities, in particular the industrial capacity.

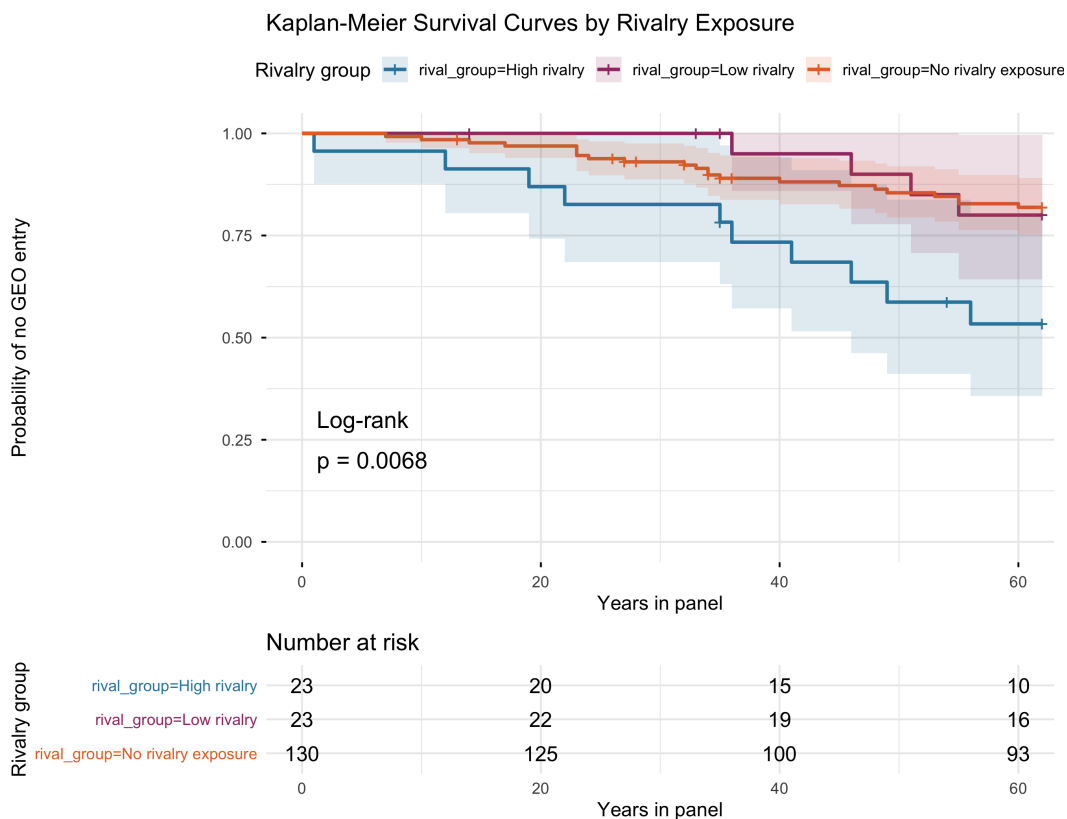


Figure 11: Conditional Kaplan-Meier survival curves on rivalry exposure. Countries with high rivalry exposure deploy into GEO much quicker than other countries, but this relation is confounded by country capacity. Log-rank test $p = 0.0068$.

Table 4 summarizes how introduction of the capacity controls affects Cox model's results. In the bivariate model, the rivalry hazard ratio is extremely significant and quite large (18.92, $p = 0.002$), but when industry shares and populations are introduced into the regression, the ratio becomes nearly unity and statistically non-significant (≈ 0.993 , $p = 0.621$). This indicates that the apparent bivariate relationship between rivalry exposure and first GEO entry is largely absorbed by basic capacity variables.

Table 4: Cox Proportional Hazards Models for First GEO Entry

Model	Specification	Rivalry HR	<i>p</i> -value	Key Controls
(1)	Bivariate	18.92	0.002	—
(2)	ln(GDP)	3.47	0.305	ln(GDP) : <i>sig.</i>
(3)	Full Controls	0.993	0.621	Industry/GDP, ln(<i>pop</i>) : <i>sig.</i>
(4)	+ Military Burden	0.908	0.149	Industry/GDP, ln(<i>pop</i>) : <i>sig.</i>

Notes: Sample of 176 countries, 9,278 country-years, 36 failures. Clustering country-level standard errors. Model (3) includes rival pressure R_{it}^{pct} , industry/GDP I_{it} , and $\ln(Pop_{it})$. Model (4) additionally includes military expenditure/GDP M_{it} .

In terms of Cox performance, the bivariate rivalry model shows quite poor prediction: the Harrell’s C statistic of 0.544 is only slightly higher than in case of coin flipping ($C > 0.5$). In the full control specification, the industrial share and country population significantly improve this statistics to 0.645. However, introduction of rivalry decreases the performance of the Cox model: its concordance drops to 0.638, and introducing military burden into the model does not change it ($C = 0.643$). Moreover, the likelihood ratio test for null hypothesis of irrelevance of rivalry in the full model does not reject the null (see Fig. 12). The full Cox models with capacity controls and without rivalry exhibit statistically identical concordance and 95% CIs overlap.

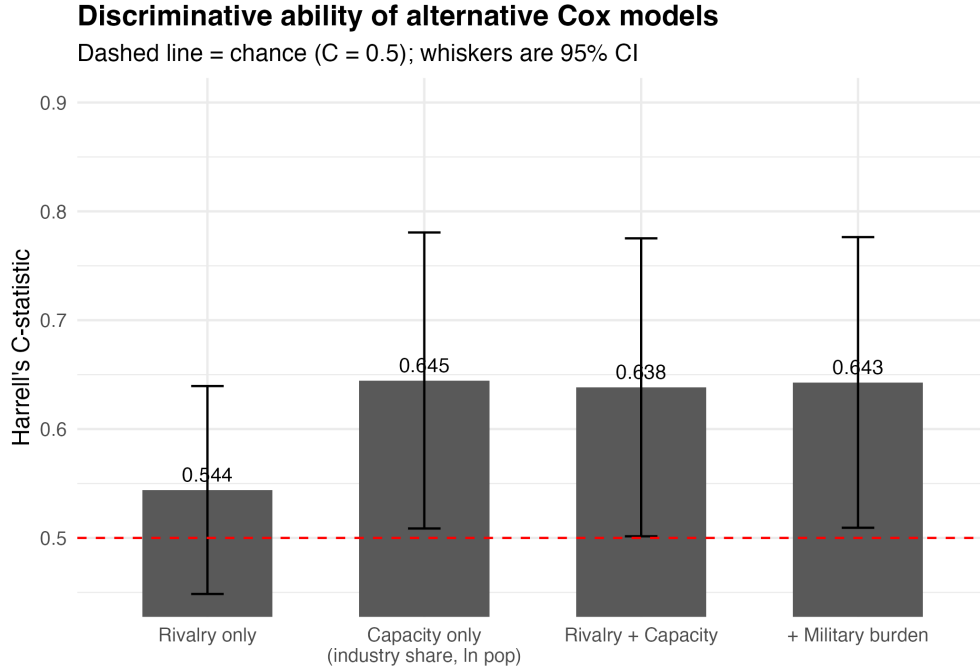


Figure 12: Alternative Cox models’ Harrell’s C -statistic for first GEO deployment. Nearly all of the Cox predictions come from controls, rivalry plays a very minor role in the prediction.

Figure 13 graphically demonstrates this feature by plotting Cox model-based predicted survival curves at various rivalry levels (0%, 1%, 10%, 40%), keeping the industry/GDP and log population

at median value (Cox Model 3). All curves overlap almost perfectly in quite a narrow confidence bounds between 85% and 93%. By comparison, in Figure 11 all survival curves exhibit clear separation, however, this difference disappears after introducing industry share and country population into Cox.

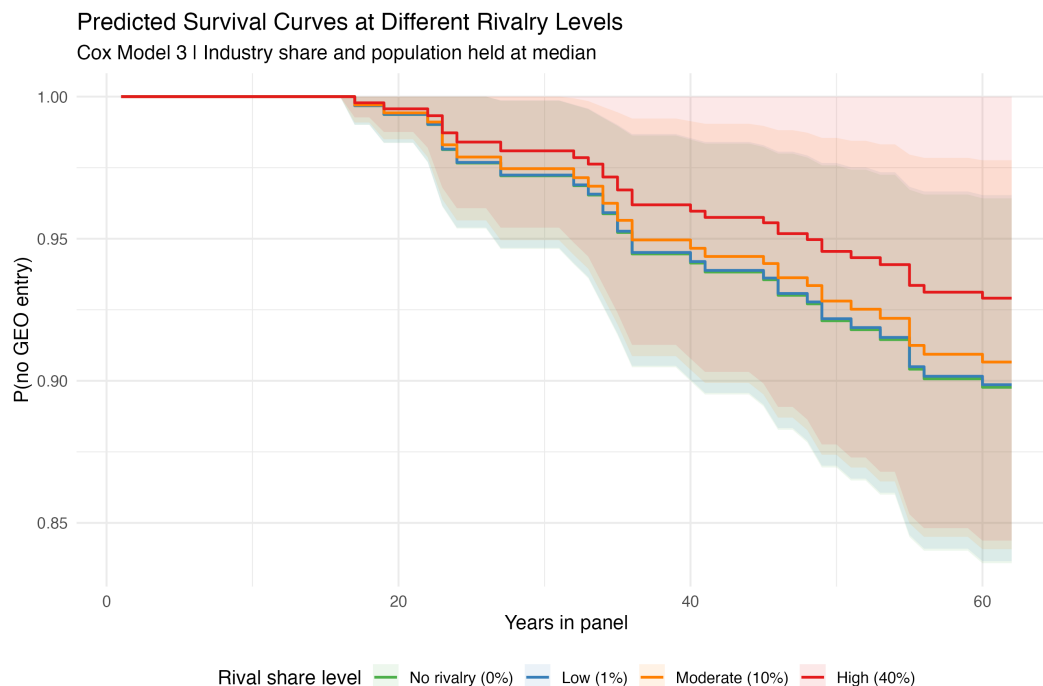


Figure 13: Predicted Cox model (Model 3) survival curves at various levels of rivalry exposure (median of industry/GDP and log population). All curves coincide almost perfectly.

Finally, Figure 14 presents the stratified Cox models' coefficients on rivalry exposure for two GDP subgroups, high-GDP and low-GDP. Both groups contain roughly equal number of units: 14 high-GDP and 15 low-GDP. In both samples the rivalry hazard ratios are statistically insignificant, and even the confidence interval bound is very tight and close to 0 in the high-GDP group and very wide in the low-GDP group (with the few observations). Low-GDP group demonstrates positive correlation of the industrial share and log-population with the entry rate.

In sum, Cox model cannot provide evidence for association with rivalry variable.

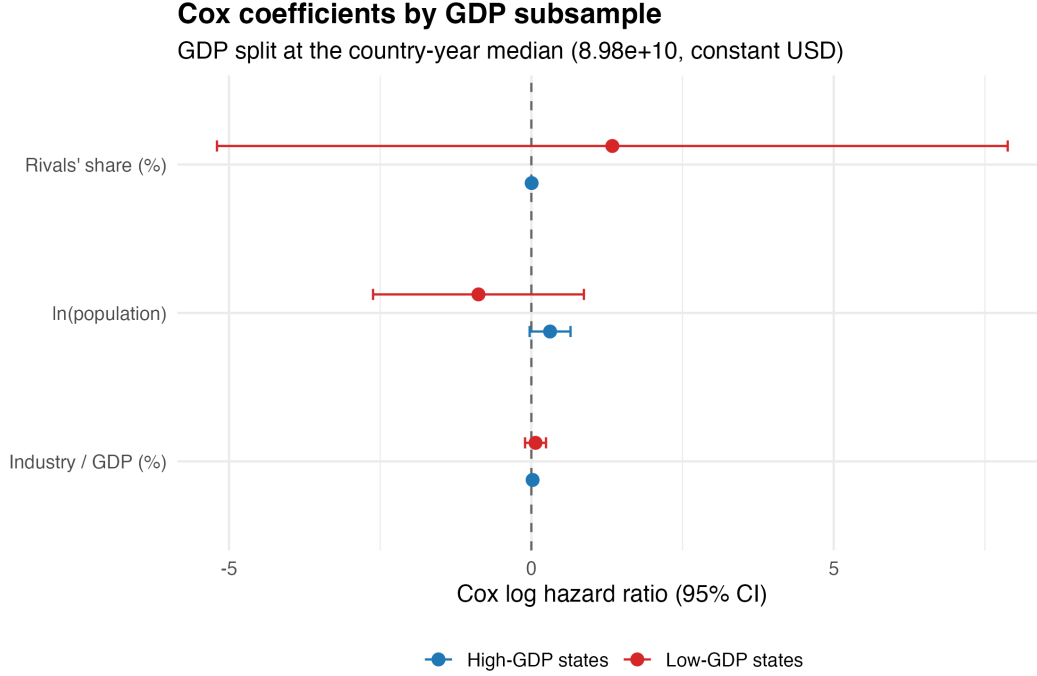


Figure 14: Cox model's hazard ratio coefficients stratified on GDP group. Rivalry effect is insignificant in both samples.

On Functional Form and Underlying Distribution of Rivalry

An alternative approach would be to consider non-linear functional form and Cox survival curve distribution. Specifically, including a squared rival-pressure term, we obtain a quite surprising coefficient estimates, which approach to statistical significance: the linear part of the coefficient is negative (significantly), but the squared rivalry term is positive (approaching significance). Together, the coefficients form a nearly significant estimate ($\hat{\beta}_R \approx -0.154, p = 0.056$; $\hat{\beta}_{R^2} \approx 0.00193, p = 0.061$). However, upon transforming the variable under inverse hyperbolic sine function, coefficients lose their statistical significance (both terms: $\hat{\beta}_R \approx -0.0559, p = 0.784$; $\hat{\beta}_{R^2} \approx 0.00184, p = 0.637$). These findings are expected taking into account that the distribution of rival pressure is heavily skewed with many zeros (at medians, 75th and 90th percentiles). Even at 99th percentile, R_{it}^{pct} equals approximately 44.2 and the kurtosis of the distribution is 97.4.